

Practice Questions-2

1. The following data give the time in months from hire to promotion to manager for a random sample of 25 software engineers from all software engineers employed by a large telecommunications firm.

5	7	229	453	12	14	18	14	14	483
22	21	25	23	24	34	37	34	49	64
47	67	69	192	125					

Calculate the mean, median, mode, variance, and standard deviation for this sample.

2. The following are the closing prices of some securities that a mutual fund holds on a certain day:

10.25	5.31	11.25	13.13	18.00	32.56	37.06	39.00
43.25	45.00	40.06	28.56	22.75	51.50	47.00	53.50
32.00	25.44	22.50	30.00	24.75	53.37	51.38	26.00
53.50	29.87	32.00	28.87	42.19	37.50	30.44	41.37

- (a) Find the mean, variance, and range for these data and interpret.
(b) Find lower and upper quartiles, median, and interquartile range. Check for any outliers.
(c) Construct a box plot and interpret.
(d) Construct a histogram.
(e) Locate on your histogram x , $x \pm s$, $x \pm 2s$, and $x \pm 3s$. Count the data points in each of the intervals $x \pm s$, $x \pm 2s$, and $x \pm 3s$ and compare this with the empirical rule.
3. A random sample of 100 households' weekly food expenditure represented by x from a particular city gave the following statistics:

$$\sum x_i = 11,000, \text{ and } \sum x_i^2 = 1,900,000.$$

- (a) Find the mean and standard deviation for these data.
(b) Assuming that the food expenditure of the households of an entire city of 400,000 will have a bell-shaped distribution, how many households of this city would you expect to fall in each of the intervals, $x \pm s$, $x \pm 2s$, and $x \pm 3s$?

4. Following data give the number of births by age group of mothers between ages 10 and 39 in a certain state in 2000. Find the median for this grouped data and interpret.

Age of mother	Number of births
10-14	895
15-19	55,373
20-24	1,22,591
25-29	1,39,615
30-34	1,27,502
35-39	68,685

5. Following data gives the distribution of the masses (in grams) of 50 salmon from a single young cohort.

Weight	155-164	165-174	175-184	185-194	195-204
Frequency	8	11	18	9	4

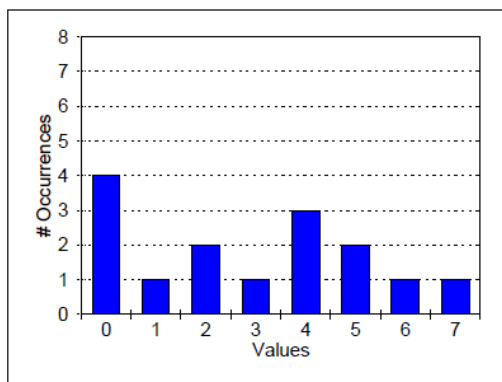
- (a) Find the mean, median, and mode.
 (b) Find range, quartile deviation, and standard deviation.

6. After a pollution accident, 180 dead fish were recovered from a stream. Following table gives their lengths measured to the nearest millimeter.

Length of fish (mm)	1-19	20-39	40-59	60-79	80-99
Frequency	38	31	59	45	7

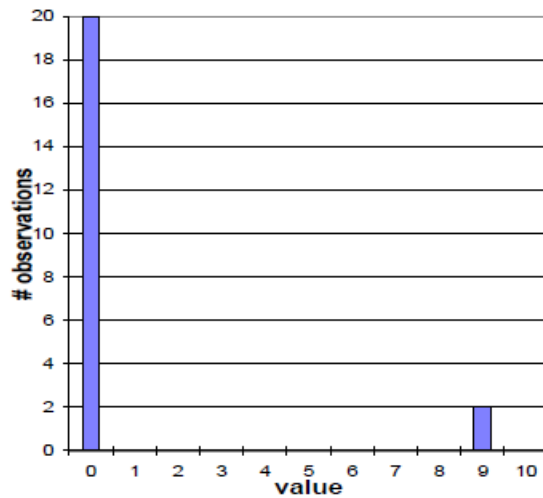
- (a) Obtain mean, median, and mode.
 (b) Obtain mean deviation about mean, median, and mode.

7. Find the median and standard deviation of the data given in the following figure:



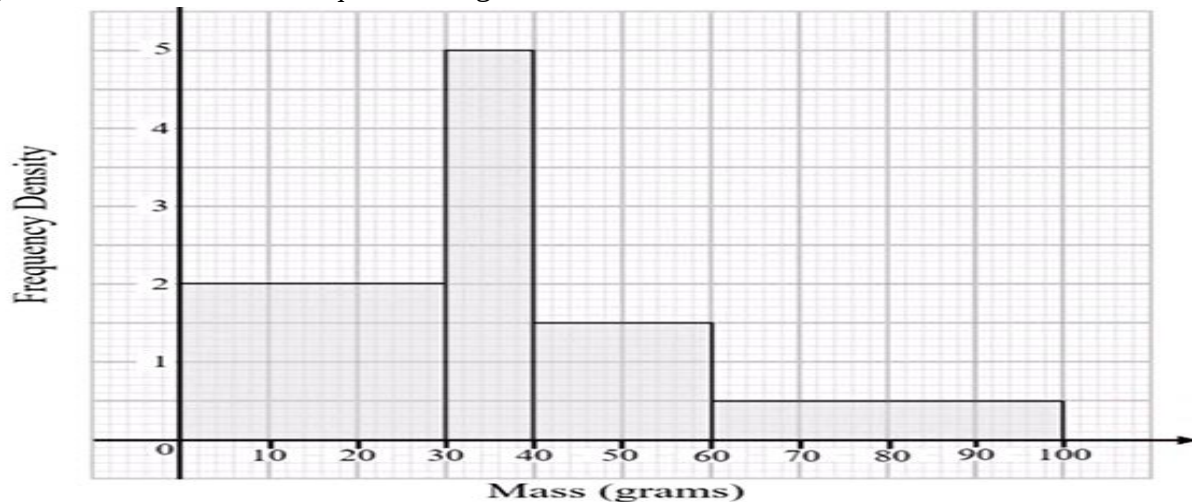
8. From the following histogram:

- (a) Obtain mean of the data.
- (b) Obtain median of the data.



9. The following histogram shows shows the masses of some potatoes.

- (a) What would be the median mass of potatoes?
- (b) What would be the lower quartile of mass of potatoes?
- (c) What would be the upper quartile of mass of potatoes?
- (d) What would be the quartile deviation?
- (e) What would be the interquartile range?



10. Five students from GLA University, took job in Data Analytics after graduation. Four of the students reported their starting monthly salaries: INR 40,000, INR 55,000, INR 55,000, INR 70,000. The fifth student did not report a starting salary. Can you tell what would be the median starting salary?